

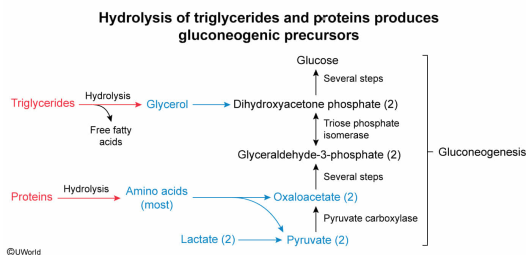
Glucose can be synthesized from several molecules present in the body. Which statement explains how glucose can be generated from both proteins and triglycerides in cells?

- ☐ A. The gluconeogenic precursor pyruvate is synthesized from amino acids in proteins or from fatty acids in triglycerides. [24%]
- ✓ ☒ B. Hydrolysis reactions release the gluconeogenic precursors glycerol (from triglycerides) and amino acids (from proteins). [31%]
- ☐ C. Acetyl-CoA, a gluconeogenic precursor, can be synthesized from both amino acids and fatty acids. [39%]
- ☐ D. Phosphogluconate and acetyl-CoA are precursors that can enter the gluconeogenic pathway. [3%]

Correct

31% answered correctly

Explanation:



Glucose and its metabolites play a central role in many biochemical reactions, and some cell types either require or exhibit a preference for glucose over other fuels. In a well-fed individual, glucose can be derived from more complex sugars and other carbohydrates in the diet. However, during prolonged periods of fasting or dietary carbohydrate restriction, the body's glucose needs must be met through the synthesis of glucose.

Gluconeogenesis is the synthesis of glucose from noncarbohydrate molecules in a pathway of reactions that resembles but is **not identical to glycolysis in reverse**. The molecules employed as gluconeogenic precursors can be metabolites of glucose produced in glycolysis (eg, pyruvate) or the **Krebs cycle** (eg, oxaloacetate), as well as certain amino acids and glycerol. In contrast, fatty acids are not gluconeogenic precursors.

The question asks for the best explanation of how glucose can be synthesized from proteins and **triglycerides**. During **proteolysis**, amino acids are released from proteins by hydrolysis reactions. Similarly, **triglyceride breakdown** by lipases **releases the glycerol** backbone through hydrolysis. Therefore, **glucose** can be **synthesized from glycerol and amino acids**, which are **gluconeogenic precursors** derived **from triglycerides and proteins**, respectively.

(Choice A) Pyruvate cannot be synthesized from fatty acids, which are not gluconeogenic precursors.

(Choice C) Acetyl-CoA cannot be used as a gluconeogenic precursor because the formation of acetyl-CoA from pyruvate is not reversible. Although acetyl-CoA can enter the Krebs cycle to eventually become oxaloacetate, the acetyl-CoA must first react with oxaloacetate to do so, and this pathway does not provide any net increase in oxaloacetate levels.

(Choice D) Neither phosphogluconate (formed in the irreversible phase of the [pentose phosphate pathway](#)) nor acetyl-CoA are gluconeogenic precursors. In addition, phosphogluconate is not produced from triglycerides or proteins.

Educational objective:

Gluconeogenesis synthesizes glucose from noncarbohydrate molecules to meet the body's glucose needs when dietary or stored carbohydrate sources are insufficient. The 3-carbon molecules lactate, pyruvate, and glycerol are gluconeogenic precursors, as are most amino acids (except for leucine and lysine).

Time spent:0 sec

Subject:
Biochemistry

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System:
Metabolic Reactions